ECON10005 Quantitative Methods 1

Tutorial in week 11: Week 10 Material

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Introduction

Zheng Fan

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- Personal website: *zhengfan.site* for some details

Don't be shy if you need help

- Discuss on Ed Discussion Board
- Attend consultation sessions: see Canvas for time and location
- Consult Stop 1, in case of special considerations,
- Contact QM-1@unimelb.edu.au for all admin issues, including group assignments.
- Send me an email: fan.z@unimelb.edu.au

Group project reminder

Data Analysis Report - Final

- Follow the instruction.
- The report is due at 10 a.m. on Monday, 25 May

Final exam

- Please note that the final exam will be pen and paper based (just like the MST).
- You will have the formula sheet and probability tables to refer to.
- Make sure that you read the information about the final exam available in the final exam module on Canvas.
- I encourage you to make use of the exam consultations when the schedule is made available.

Final exam

- Try to start your revision as early as you can.
- Reviewing the lecture slides and re-attempting the pre-tutorial and tutorial questions is a good idea.
- Rather than passively reading through solutions, try to practice attempting questions **by hand**, and then check the solutions. From my own personal experience, you will recall information much better this way.
- It is a good idea to try and complete the practice exam under timed exam conditions if you can.

Glossary

- Before we discuss this week's tutorial questions, we will briefly review the glossary of simple linear regression provided in the lecture slides for week 10.
- As was the case for one-sample and two-sample statistical inference, we need to distinguish the **population** and the **sample** in the context of simple linear regression.

Linear regression

Linear Regression: Estimation

Linear regression model: $E\left(Y_i \mid X_i\right) = \beta_0 + \beta_1 X_i$

How to estimate β_0 and β_1 (the line that best describes the relationship)?

Linear Regression: Estimation

Linear regression model: $E\left(Y_i \mid X_i\right) = \beta_0 + \beta_1 X_i$

How to estimate β_0 and β_1 (the line that best describes the relationship)?

Minimise the sum of squared errors

$$\text{minimize} \qquad \text{SSE} = \sum_{i=1}^{n} \hat{U}_i^2 = \sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i \right)^2$$

yields

$$\hat{\beta}_1 = \frac{s_{XY}}{s_X^2} \qquad \text{and} \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

where

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)$$

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2$$

Question 1

Recall that you have two samples of data for quantity supplied and quantity demanded given various prices. The two population regression models are:

- We assumed a linear regression for the quantity supplied (QS) by this particular market as a function of price (P). $QS_i = \beta_0 + \beta_1 P_i + U_i$
- We assumed a linear regression for the quantity demanded (QD) by this particular market as a function of price (P). $QD_i = \beta_0 + \beta_1 P_i + U_i$

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- We assumed a linear regression for the quantity demanded (QD) by this particular market as a function of price (P). $QD_i = \beta_0 + \beta_1 P_i + U_i$

Using your answers for the pre-tutorial 10, question 1, 2, discuss the following questions.

1. Write down the estimated regression lines for quantity demanded and quantity supplied.
2. Interpret the two coefficients on P and discuss what these mean in explaining the quantity supplied and the quantity demanded.
3. Using the SSE values you estimated for the quantity demand and quantity supply functions, discuss which equation provides a better fit to the observed data. Explain briefly.

Question 1

1. Write down the estimated regression lines for quantity demanded and quantity supplied.

The estimated regression lines are:

$$\widehat{QS}_i = 6.898 + 1.037P_i$$
$$\widehat{QD}_i = 2.644 - 0.120P_i$$

Question 1

2. Interpret the two coefficients on P and discuss what these mean in explaining the quantity supplied and the quantity demanded.

Supply Function:

- The coefficient on P is 1.037.

How do we interpret this coefficient?

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How do we interpret this coefficient?

A positive coefficient indicates that when the price increases, the quantity supplied will also increase.

This estimate suggests that with every $1 increase in price, the quantity supplied increases by 1.037 tonnes on average.

Question 1

Demand Function:

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These results align with economic theory: there is a positive relationship between price and quantity supplied, and an inverse relationship between price and quantity demanded.

Question 1

3. Using the SSE values you estimated for the quantity demand and quantity supply functions, discuss which equation provides a better fit to the observed data. Explain briefly.

- SSE for the supply equation: 92.6610
- SSE for the demand equation: 0.0217

The quantity demand equation provides a much better fit to the observed data because it has a substantially smaller SSE value.

Since SSE measures the total squared prediction errors, a smaller SSE indicates that the predicted values are much closer to the observed data values.

Therefore, the quantity demand function fits the observed data considerably better than the quantity supply function.

Question 1

3. Using the SSE values you estimated for the quantity demand and quantity supply functions, discuss which equation provides a better fit to the observed data. Explain briefly.

Important Note

A smaller SSE does not always imply a better model when different datasets or variables are used.

- A more commonly used measure is R^2, which reports the proportion of variation in the dependent variable explained by the model and is scale-free.

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

- I would personally suggest you use R^2, as it measures 'goodness of fit'.

Question 2

Refer to the pre-tutorial 10, question 3.

Recall that the spreadsheet Funds.xlsx contains data on the performance of 157 investment funds, with their annual Return (Y) and Risk (X), reported in percentages. Finance theory suggests that Risk and Return are positively related, meaning riskier funds should offer higher average returns.

Question 2

Refer to the pre-tutorial 10, question 3.

Recall that the spreadsheet Funds.xlsx contains data on the performance of 157 investment funds, with their annual Return (Y) and Risk (X), reported in percentages. Finance theory suggests that Risk and Return are positively related, meaning riskier funds should offer higher average returns.

1. Write down the estimated regression line.
2. What does this intercept represent in the context of investment funds?
3. Interpret the coefficient on X and discuss what it means in explaining the annual return. Does this agree with the finance theory?

Let's spend 5 minutes discussing this in groups.

Question 2

1. Write down the estimated regression line.

The estimated regression line is:

$$\widehat{Return}_i = -0.611 + 0.957\,\text{Risk}_i$$

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In this regression, the intercept is –0.611.

How do we interpret this coefficient?

If a fund had zero risk, the model predicts its average return would be –0.611%.

Since funds with exactly zero risk don't really exist, this number mainly acts as the line's starting point. It ensures the regression line fits the data correctly, but it doesn't have much practical financial meaning.

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- The coefficient on Risk is 0.957.
- A positive coefficient indicates that there is a positive relationship between the risk and returns.
- This estimate suggests that every one per cent increase in risk is associated with an increase of return by 0.957 percent on average.

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Do these results align with the finance theory?

Yes, there is a positive relationship between risk and the returns on an investment.

Any final questions?

Thanks for your attention! 😃

Let me know if you have any questions.